

Lecture 10

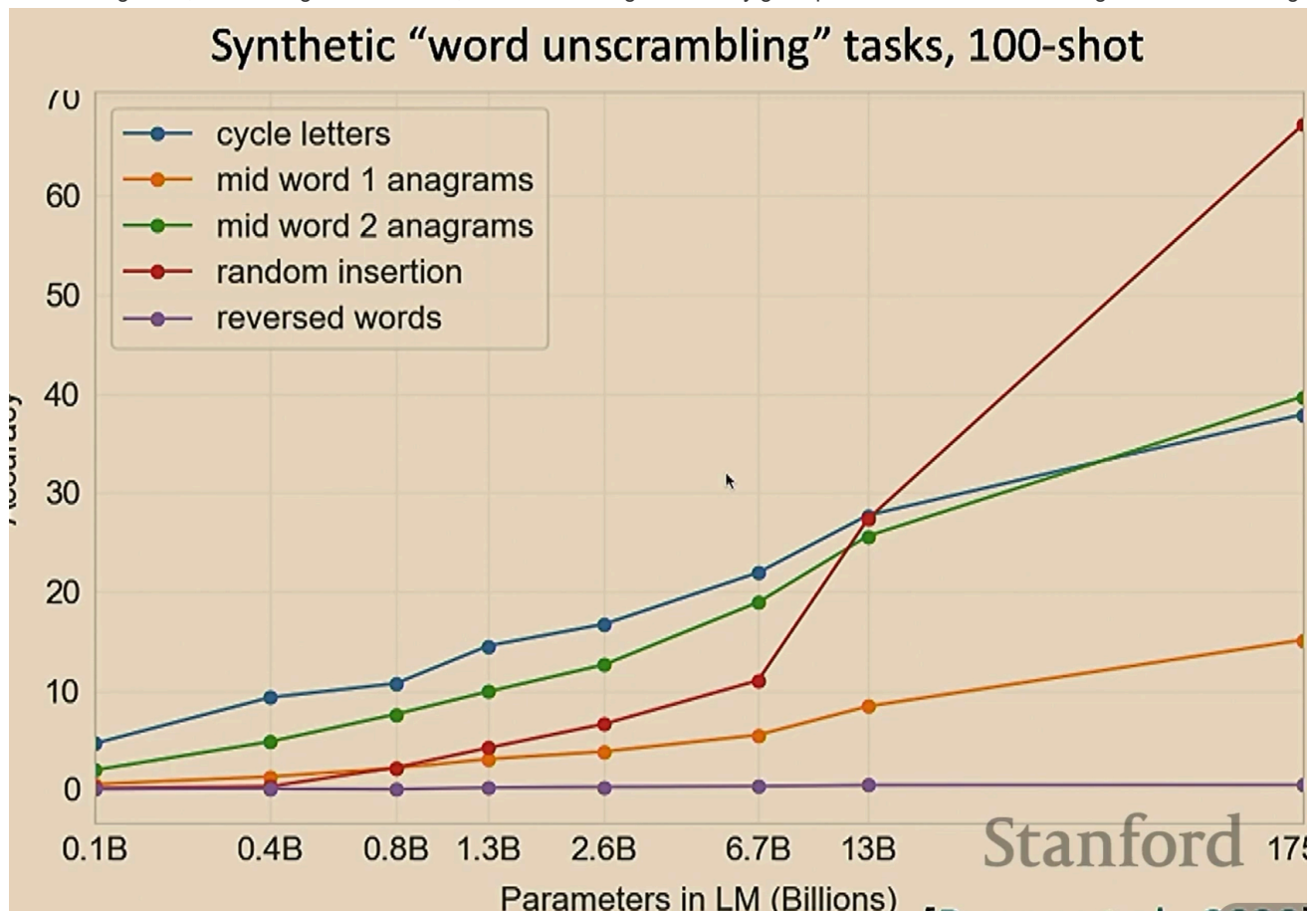
note: this lecture notes are from the spring 2024 lecture, because on the schedule i found it to be covering more recent algos that are used. (DPO)

World Models: its a model of how it changes the world/environment, when its agents are taking actions

Post-training: From predicting 1 or 2 token, to LLM like GPT

1. Zero-shot and Few shot In context learning

- **Natural language inference task** : missed from last lecture, which is given 2 sentences and the model is supposed to give out the correlations between them.
- from GPT1 to GPT2, params from 117M to 1.5B, trained data from 4GB text to 40GB (links from reddit, webtext)
- **zero-shot learning**
 - just querying the model, letting it do tasks which it was not trained for, with no examples, and no gradient updates.
 - **we can do it by**
 - question answering, through given text.
 - classifying the probability of different sequences, like which object "it" refers to in a sentence.
 - this method used by GPT, beats the SoTA on benchmarks
- then GPT scaled to 175B, data of 600GB, and based on **Few shot learning**
- **few shot learning**
 - refers as **in-context learning** (ICL). Before asking the model to perform specific tasks, you provide few examples of the task. And still the same, there are **no gradient updates**
 - in terms of doing tasks, with the higher model size, few shot learning shows very good performance in tasks of doing word unscrambling.



- similarly for creative tasks, there are no right/wrong answer
- and token level mistakes are penalized equally by the model, but different mistakes have different importance
- the results for the instruction pair provided by the human may not be always correct, especially in complex tasks
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3. Optimizing for human preferences(DPO/RLHF)

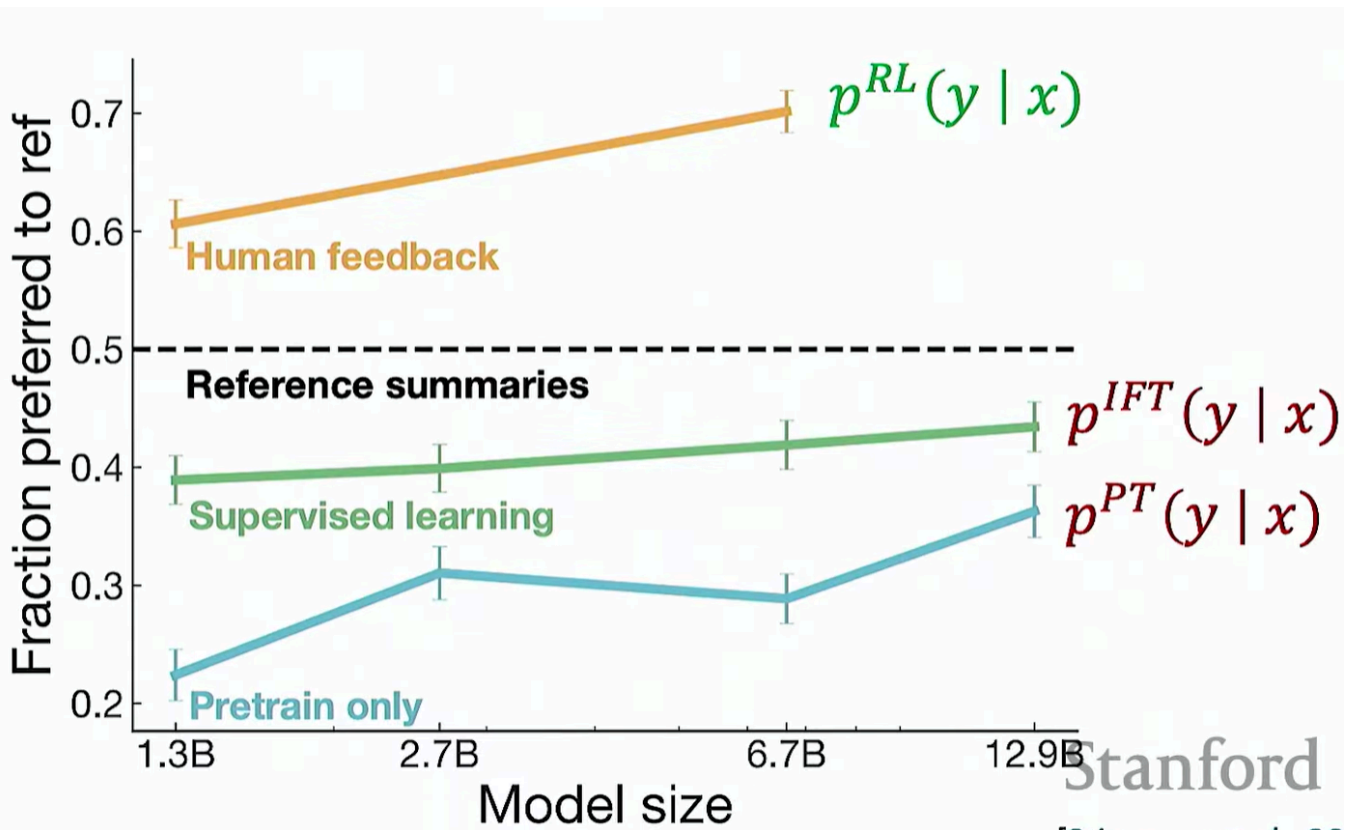
- so on a specific task, you can train the model to be more specified towards human preferences.
 - so this can be done through RL, where we assign rewards to the outcomes from the task and we optimize it

• RLHF pipeline

- Firstly, we instruction tune a **pretrained model**, so it could be getting to completing tasks.
- Then we train a **reward model**, in which it responds in a way like how much would a human like/hate this response
- We need to optimize the policy by training against the reward model, to maximize the expectancy.
 - How to get the rewards:
 - instead of human labelling the data, we train a model simulating human preferences
 - However, human judgements are noisy
 - Solution: instead of asking for the exact number of scores, you ask for a comparison between several responses, ranking them.
- **Optimizing the Reward model**
 - Bradley Terry comparison model
 - $$J_{RM}(\phi) = -\mathbb{E}_{(x,y^w,y^l)\sim D} [\log \sigma (RM_\phi(x,y^w) - RM_\phi(x,y^l))]$$
 - w represent winning sample, l represents losing, letting the model learn the good ones and the bad ones

• Optimizing the language model/policy

- $$\mathbb{E}_{\hat{y}\sim p_\theta^{RL}(\hat{y}|x)} [RM_\phi(x,\hat{y})]$$
 - this is the objective expectation that we want to optimize with the input sequence x , and the output sequence $\hat{y}$.
 - however, learning has wrong, inadequate responses, with given high rewards, this just leads to unevitable situations where you can't fix the model back again.
- **Solve**
 - we can add penalty to the model when it goes too far away from its initialization
- $$\mathbb{E}_{\hat{y}\sim p_\theta^{RL}(\hat{y}|x)} \left[RM_\phi(x,\hat{y}) - \beta \log \left(\frac{p_\theta^{RL}(\hat{y} | x)}{p^{PT}(\hat{y} | x)} \right) \right]$$
 - this referred as the **KL divergence**
 - PT referred to the pretrained model
 - and in the case of $p_\theta^{RL}(\hat{y} | x) > p^{PT}(\hat{y} | x)$, where if the current RL model gives too high prob distribution to an output sequence compared to the pretrained model, then the penalty would be higher.
- **Eventually we optimize through RL Algos** - increasing the probability of the distribution function giving higher probabilities for high reward responses, and training, updating the distribution function.... maximizing the expectancy objective
- **Performance**



- FOR human Preferences based on text summaries generated through the different trained models

4. DPO (Alternative to RLHF)

- we don't separately learn a **reward model** anymore and we can represent it in terms of the **language model** p_θ or **policy**
- Linkage between the log probabilities from the language model with the human preferences
- Derivations**

$$\mathbb{E}_{\hat{y} \sim p_\theta^{RL}(\hat{y}|x)} \left[RM_\phi(x, \hat{y}) - \beta \log \left(\frac{p_\theta^{RL}(\hat{y}|x)}{p^{PT}(\hat{y}|x)} \right) \right]$$

- this is the closed solution (can directly see the optimum) for the equation above, it is from the boltzman distribution, and $Z(x)$ is the normalizing constant.

$$p^*(\hat{y}|x) = \frac{1}{Z(x)} p^{PT}(\hat{y}|x) \exp \left(\frac{1}{\beta} RM(x, \hat{y}) \right)$$

- rearrange

$$RM(x, \hat{y}) = \beta \log \frac{p^*(\hat{y}|x)}{p^{PT}(\hat{y}|x)} + \beta \log Z(x)$$

- now, this shows that the language model implicitly defines a reward model, although it may not be the optimal one.
- and initially the p^{RL} model is just the pretrained model

$$RM(x, \hat{y}) = \beta \log \frac{p_\theta^{RL}(\hat{y}|x)}{p^{PT}(\hat{y}|x)} + \beta \log Z(x)$$

- now we fit it back to the **Bradley-Terry reward optimization model**
- whats magical here is that because we are computing the difference, so the partition function $Z(x)$ which is hard to compute would be cancelled out

$$RM_\theta(x, y^w) - RM_\theta(x, y^l) = \beta \log \frac{p_\theta^{RL}(y^w|x)}{p^{PT}(y^w|x)} - \beta \log \frac{p_\theta^{RL}(y^l|x)}{p^{PT}(y^l|x)}$$

- leading to

$$J_{DPO}(\theta) = -\mathbb{E}_{(x, y^w, y^l) \sim D} [\log \sigma (RM_\theta(x, y^w) - RM_\theta(x, y^l))]$$

- **RL + modelling:**

- still has problems of reward hacking
- but achieved peak performance in cases like alphago, which is in closed, well-defined environments.